Instalacija radnog okruženja

Branimir Ćaran

25.3.2025.| Regionalni centar izvrsnosti za robotske tehnologije

Sadržaj

- Instalacija *VMWare*-a
- Instalacija *Linux Ubuntu 22.04*
- Korištenje *Linus Ubuntu* OS
- Instalacija *ROS2 Humble*
- Pokretanje *ROS2* programa

crta

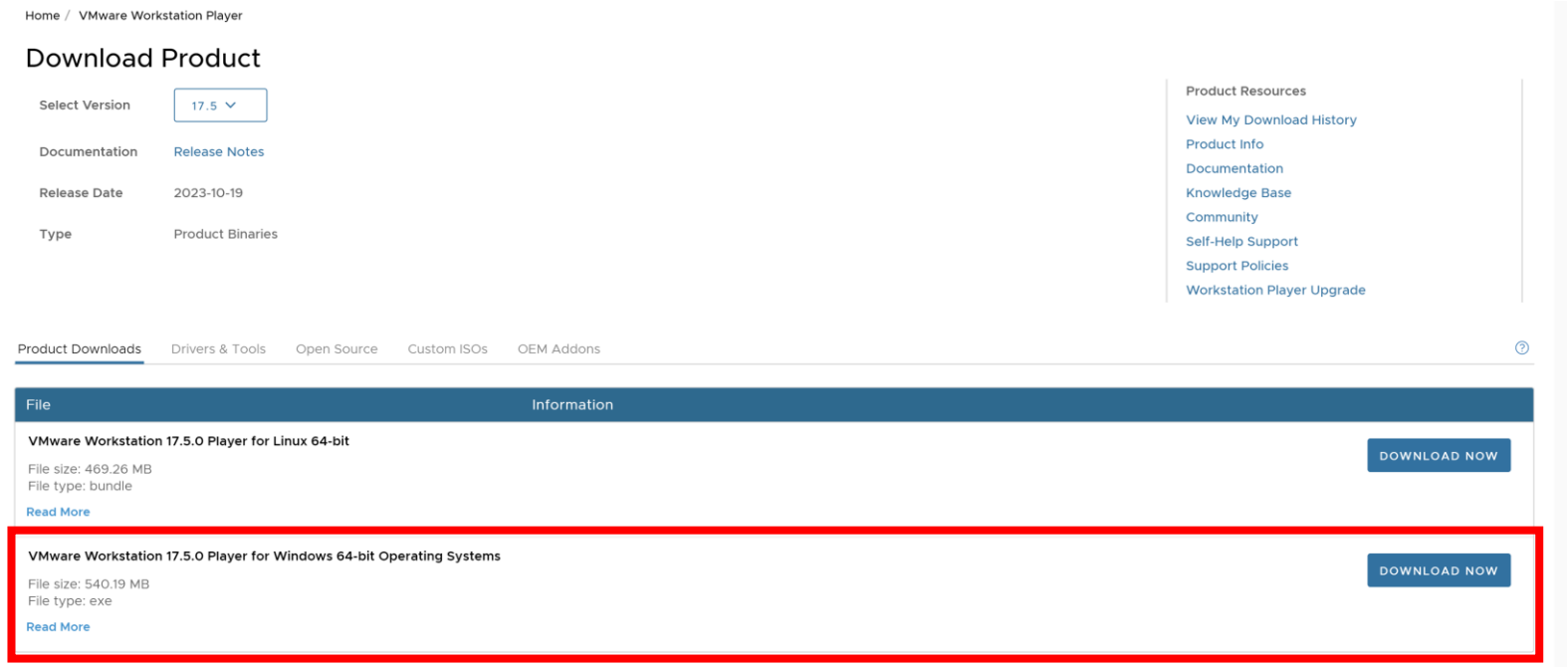
Instalacija *Linux Ubuntu 22.04*

- Moguće je pratiti prezentaciju *step by step* ili pogledati kompilaciju videa:

1. Instalacija *VMWare* programa – video
   a) Broadcom je kupio VMWare – moguće da će biti potrebe se ulogirati u stranicu
2. Instalacija *Linux Ubuntu 22.04* – video
3. Instalacija *ROS2 Humble* – video
4. Instalacija *Visual Studio* Code-a - video
5. Podešavanje *Visual Studio Code*-a - video

Instalacija *VMWare*-a

- Program za pokretanje virtualnih mašina na postojećem računalu
- Potrebno je preuzeti s linka i instalirati



Preuzimanje *Linux Ubuntu 22.04*

- Potrebno je preuzeti *.iso* „sliku” *desktop* operativnog sustava s linka

Ubuntu 22.04.3 LTS (Jammy Jellyfish)

Select an image

Ubuntu is distributed on three types of images described below.

Desktop image

The desktop image allows you to try Ubuntu without changing your computer at all, and at your option to install it permanently later. This type of image is what most people will want to use. You will need at least 1024MiB of RAM to install from this image.

64-bit PC (AMD64) desktop image

Choose this if you have a computer based on the AMD64 or EM64T architecture (e.g., Athlon64, Opteron, EM64T Xeon, Core 2). Choose this if you are at all unsure.

Server install image

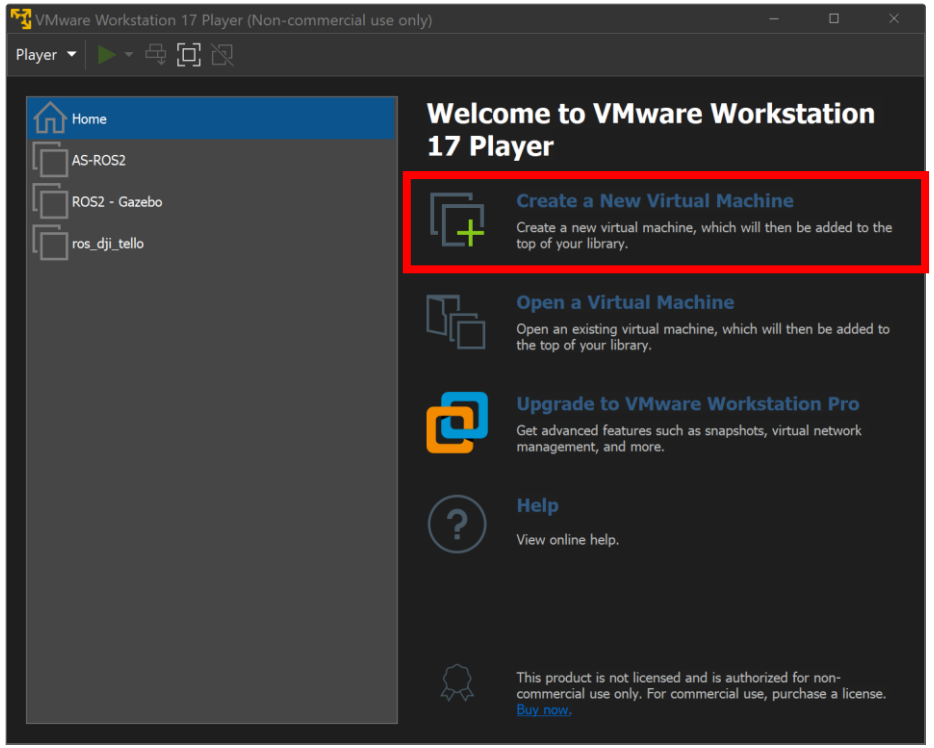
The server install image allows you to install Ubuntu permanently on a computer for use as a server. It will not install a graphical user interface.

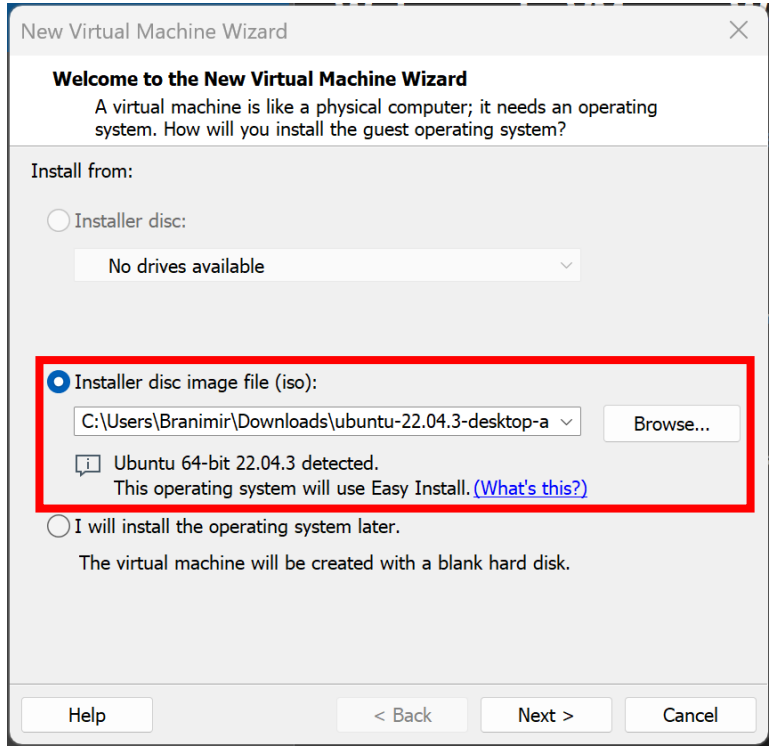
64-bit PC (AMD64) server install image

Choose this if you have a computer based on the AMD64 or EM64T architecture (e.g., Athlon64, Opteron, EM64T Xeon, Core 2). Choose this if you are at all unsure.

Instalacija *Linux Ubuntu 22.04*

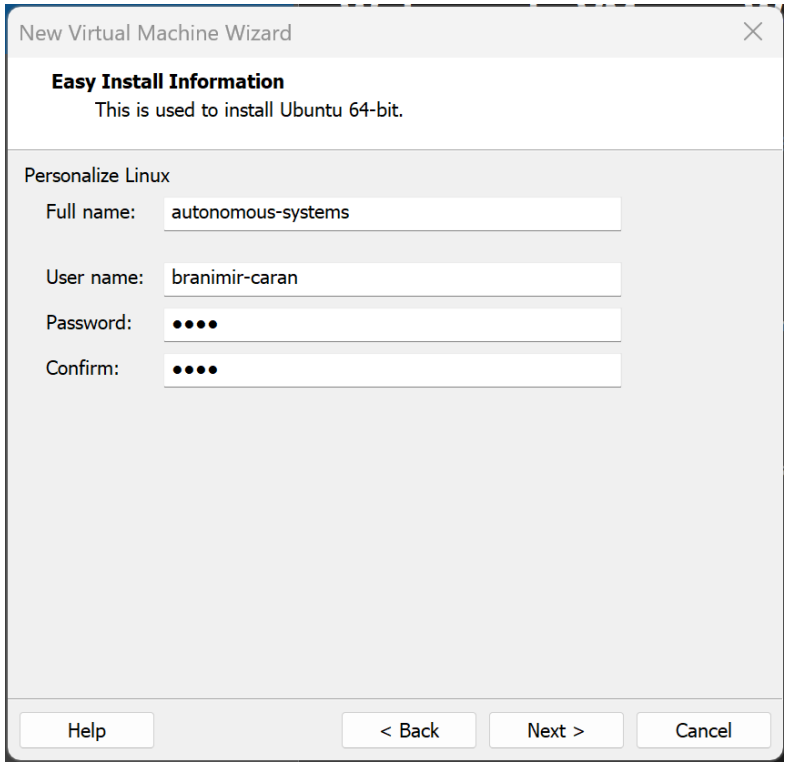
- Pokrenuti *VMWare* i stvoriti novu virtualnu mašinu i odabrati putanju do *.iso* slike preuzetog *Linux Ubuntu 22.04* OS-a





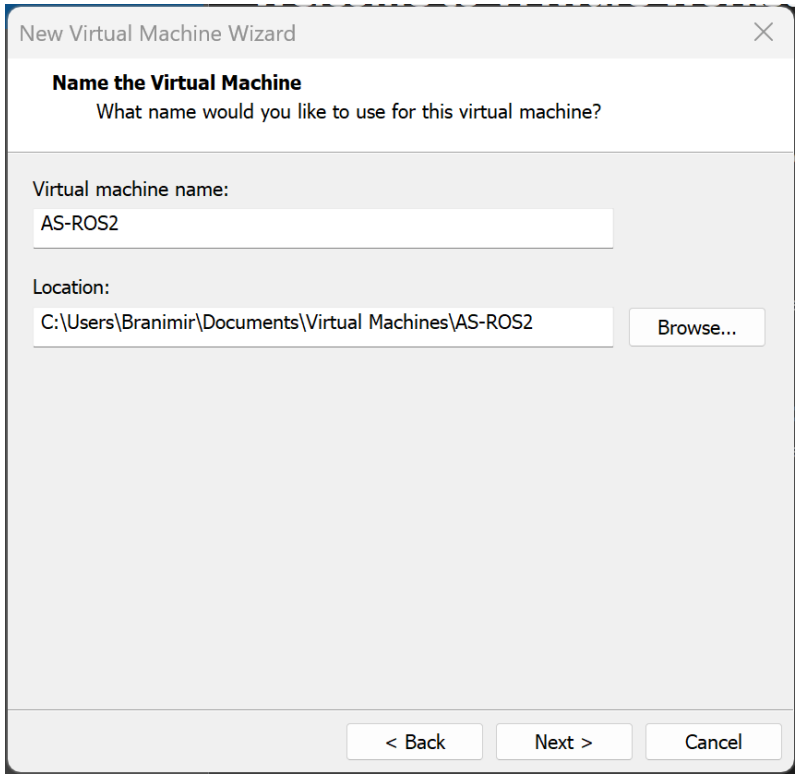
Instalacija *Linux Ubuntu 22.04*

- Potrebno je ispuniti detalje o virtualnoj mašini
- **VAŽNO**: *User name* – mora sadržavati vaše ime-prezime kako bi vas mogli raspoznati prilikom pregleda zadaća
- *Password* – staviti kratku šifru, jer ćemo je često koristiti (npr. *ros2*)



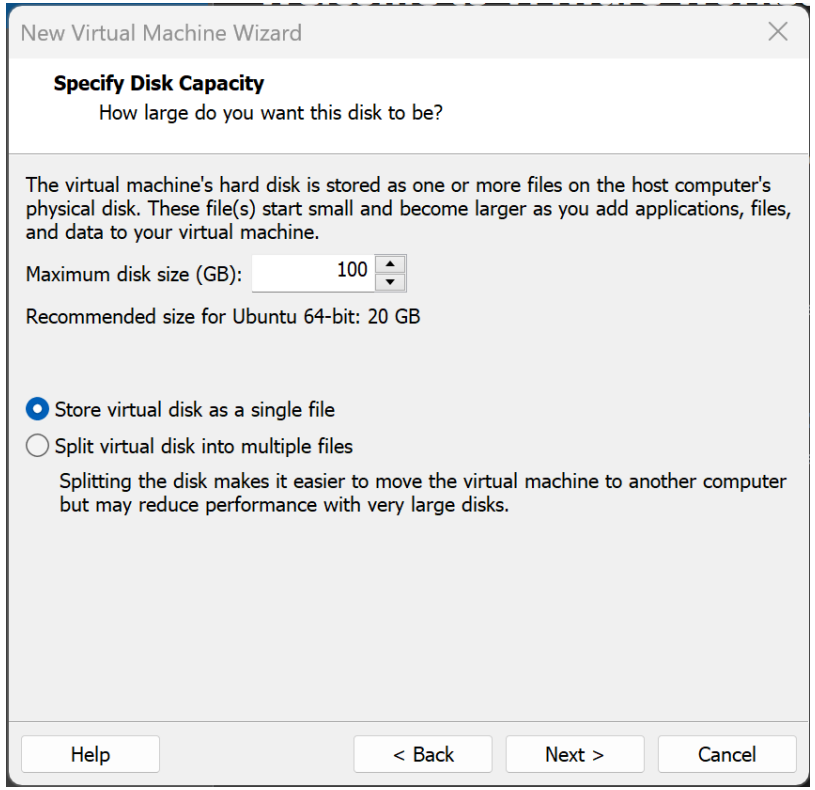
Instalacija *Linux Ubuntu 22.04*

- Odabrati kako će se zvati i gdje će se spremiti virtualna mašina



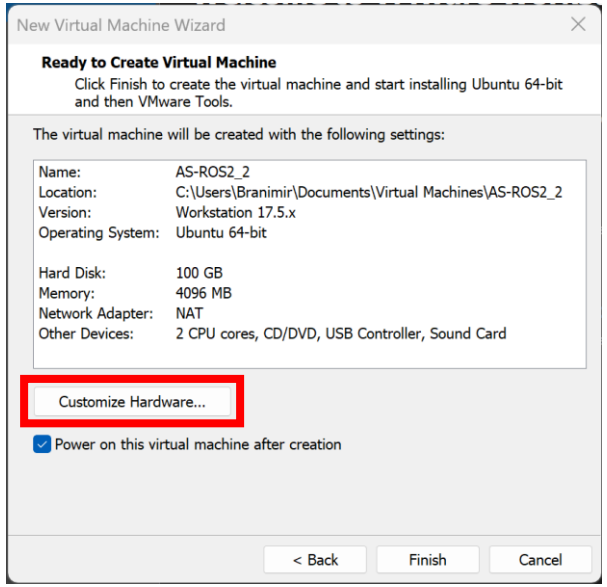
Instalacija *Linux Ubuntu 22.04*

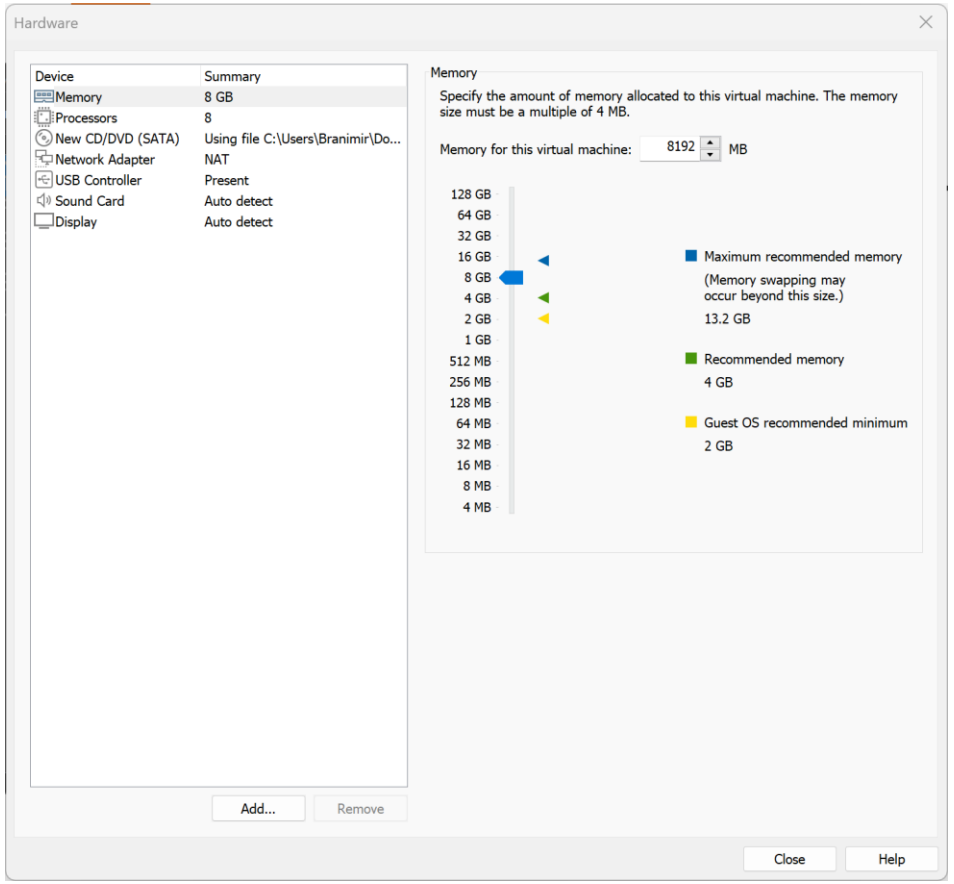
- Podesiti memoriju i način konfiguriranja diska – preporuka je definirati 100GB memorije



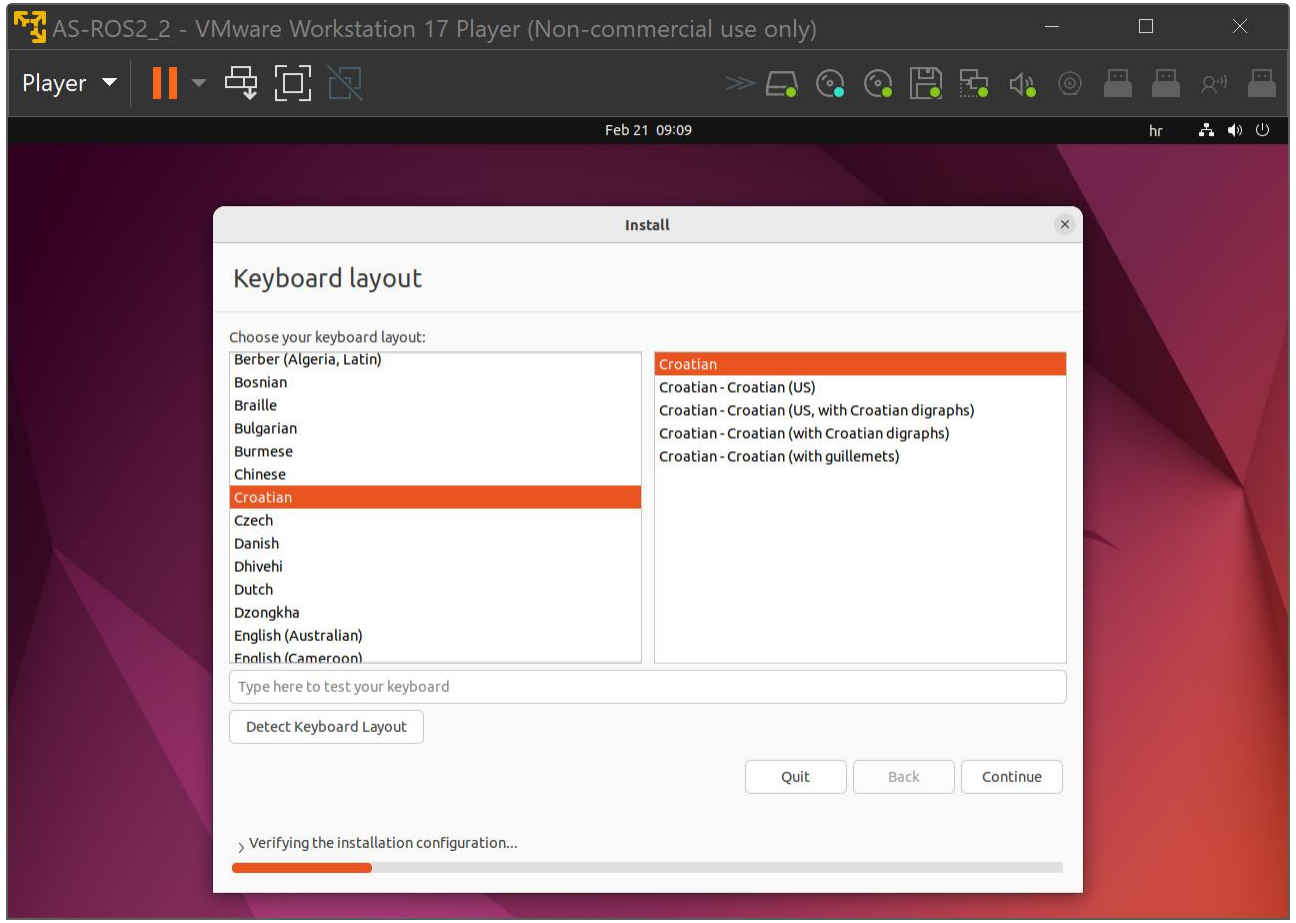
Instalacija *Linux Ubuntu 22.04*

- Podesiti performanse virtualne mašine
- RAM memorija – 8GB
- Broj jezgri: 8

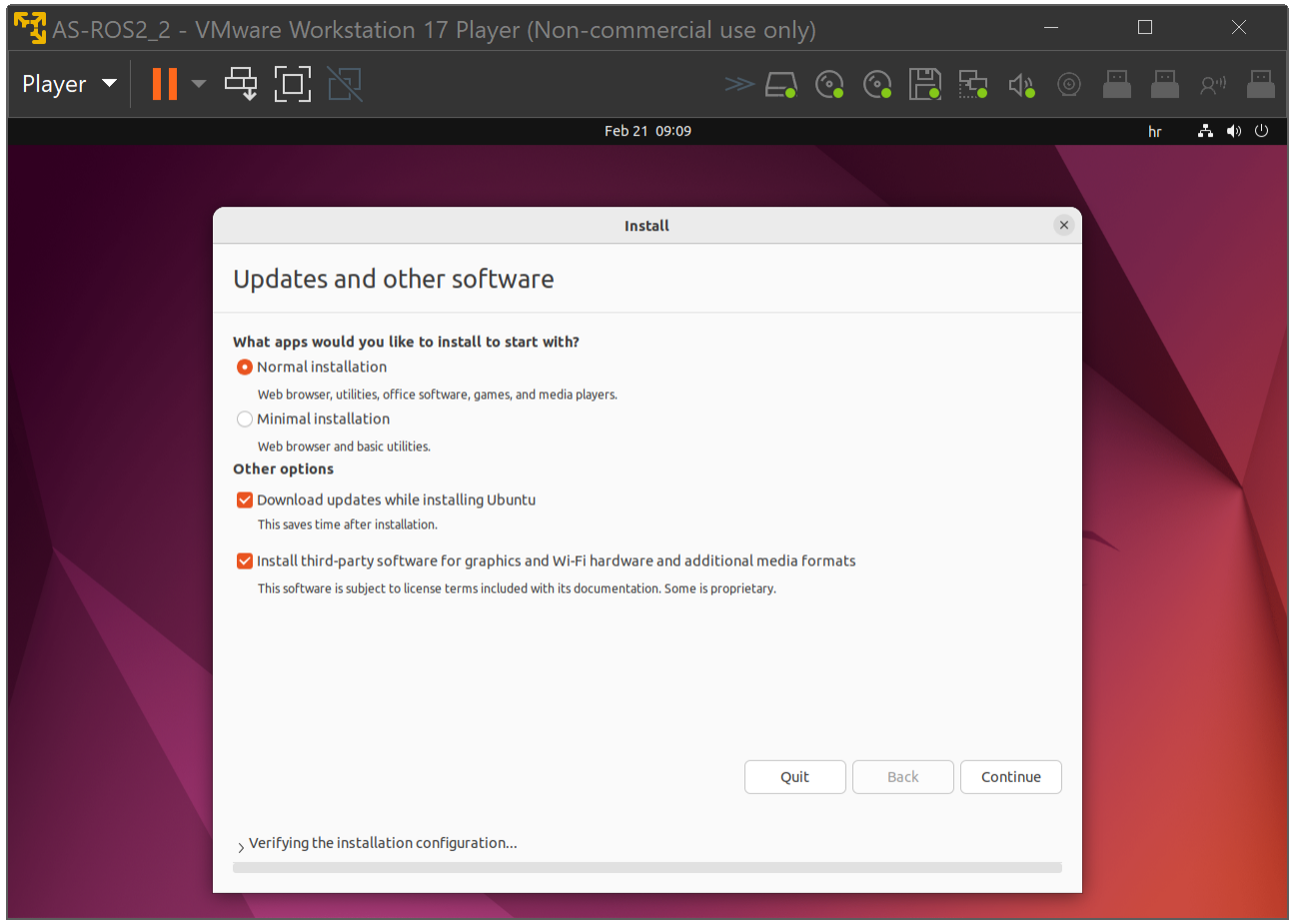




Odabrati hrvatsku tipkovnicu

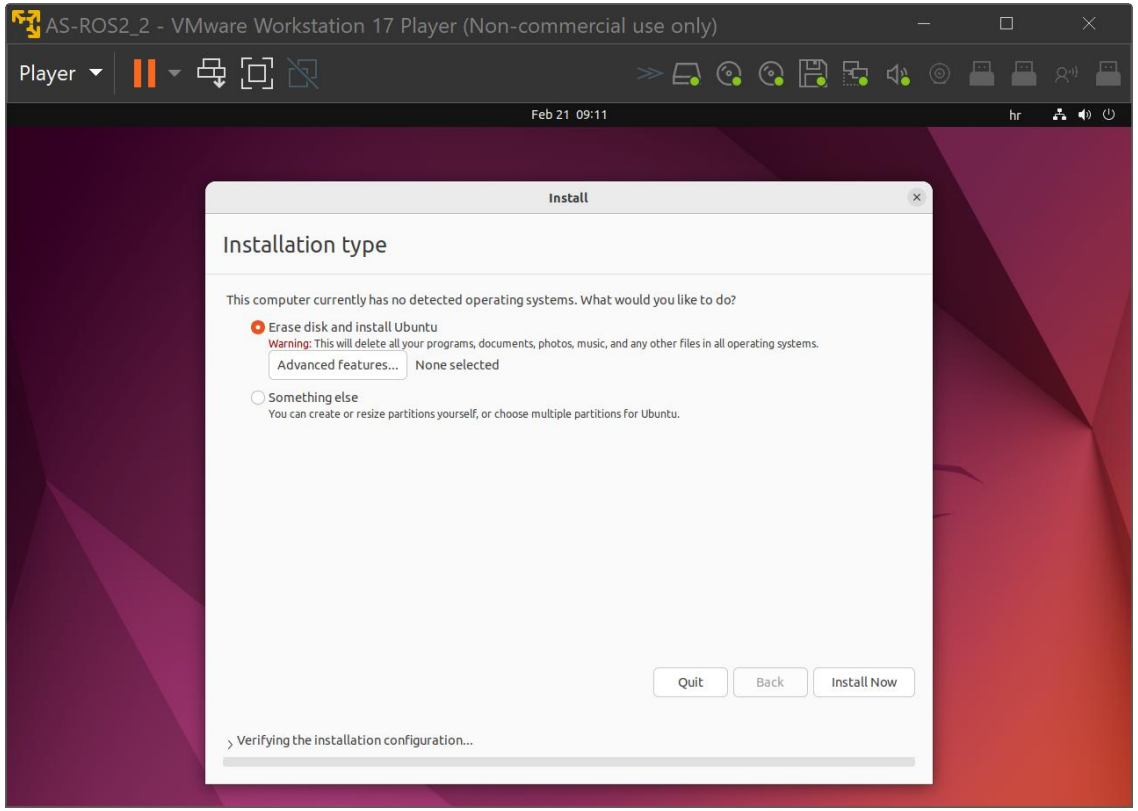


Podesiti instalacije i nadogradnje



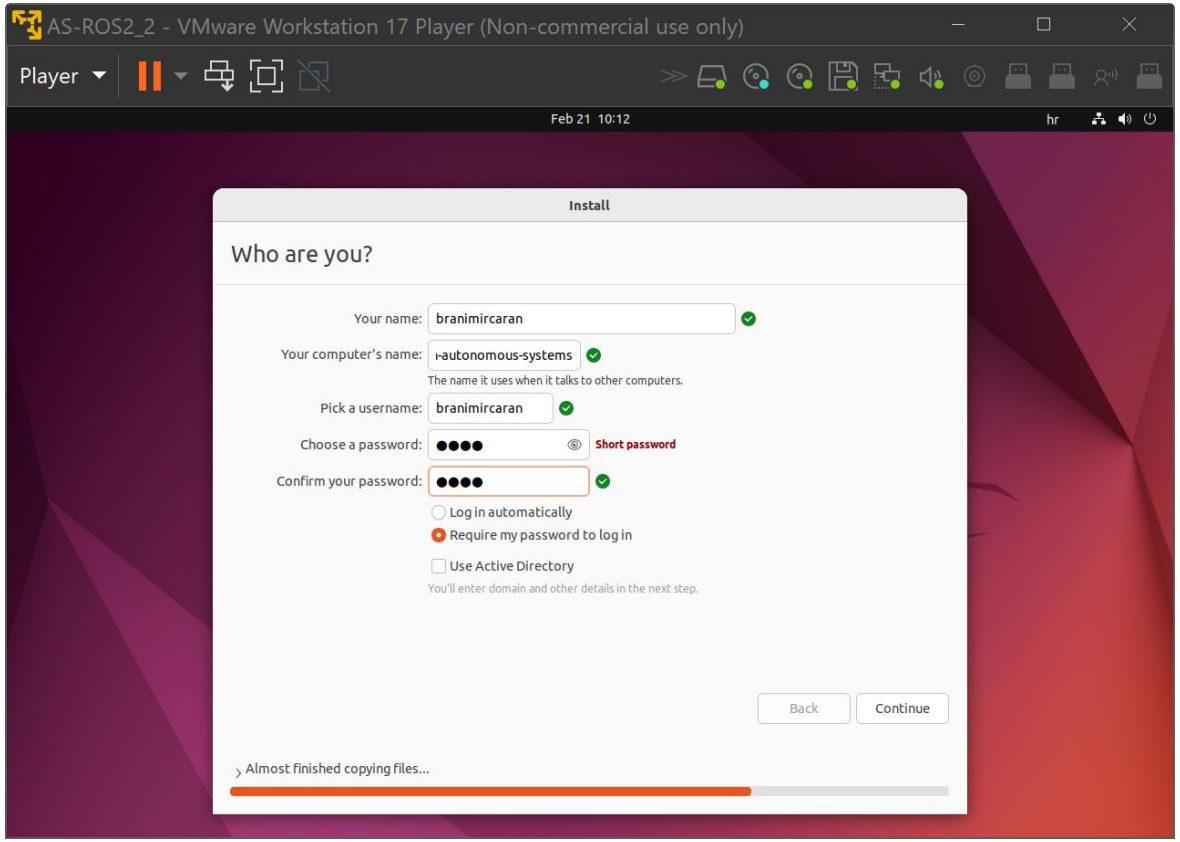
Izbrisati virtualni disk

- Briše se virtualnih 100GB koje su ionako već prazne



Dovršetak instalacije

- Popuniti na sličan način kao i u prethodnim slajdovima



Instalacija *ROS2 Humble*

- Potrebno je pratiti upute na linku za instalaciju *ROS2 Humble*
- Obratiti pozornost da se instalira *ros-humble-desktop* zbog svih potrebnih alata za simulaciju i vizualizaciju
- Obavezno instalirati *ros-dev-tools*

Desktop Install (Recommended): ROS, RViz, demos, tutorials.

```
sudo apt install ros-humble-desktop
```

ROS-Base Install (Bare Bones): Communication libraries, message packages, command line tools. No GUI tools.

```
sudo apt install ros-humble-ros-base
```

Development tools: Compilers and other tools to build ROS packages

```
sudo apt install ros-dev-tools
```

Instalacija *ROS2 Humble*

- Za provjeru potrebno je pokrenuti naredbe s kraja uputa

Try some examples

Talker-listener

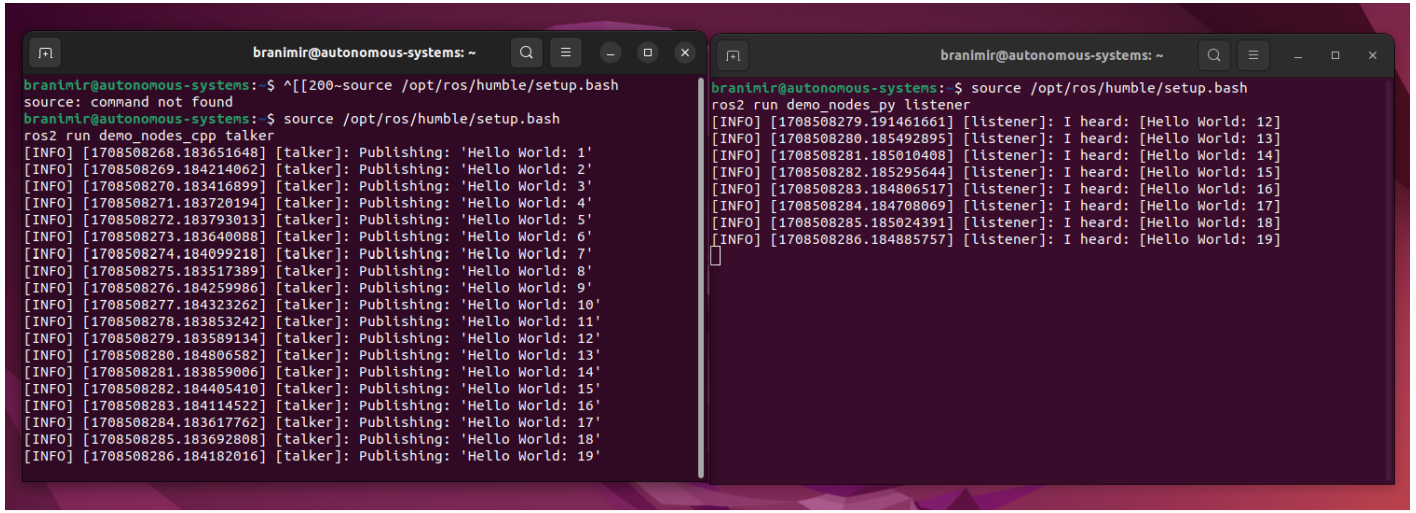
If you installed `ros-humble-desktop` above you can try some examples.

In one terminal, source the setup file and then run a C++ `talker`:

```
source /opt/ros/humble/setup.bash
ros2 run demo_nodes_cpp talker
```

In another terminal source the setup file and then run a Python `listener`:

```
source /opt/ros/humble/setup.bash
ros2 run demo_nodes_py listener
```



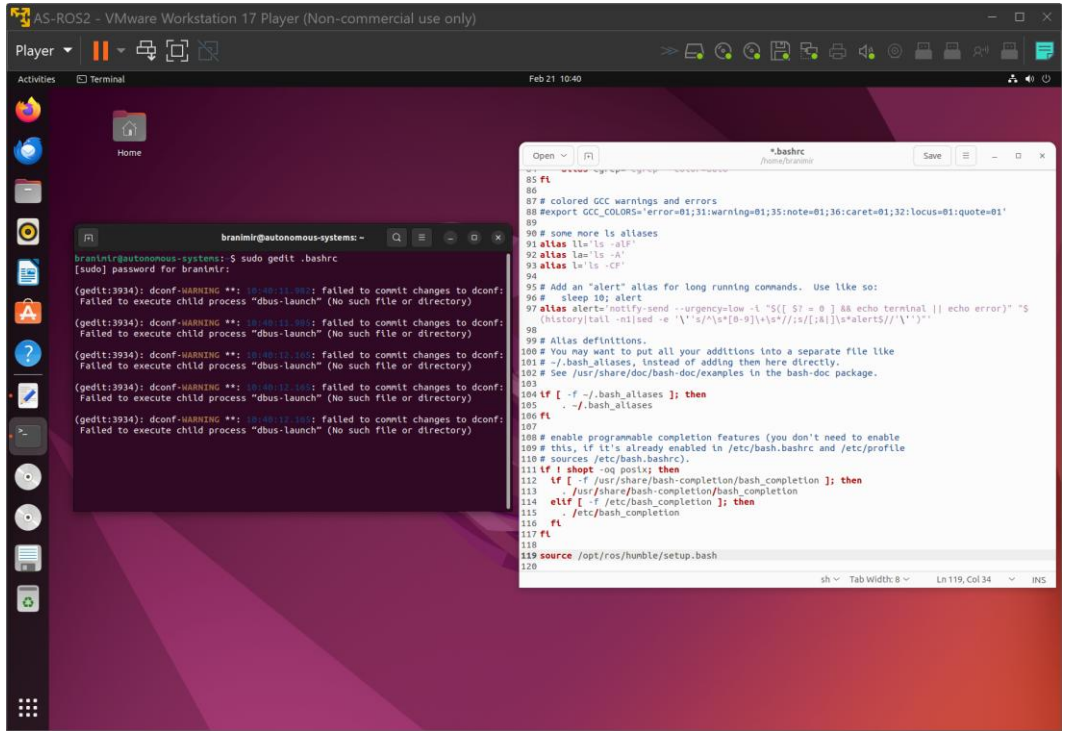
Preporuka

- Preporuka je dodati u *.bashrc* naredbu za *source*-anje

```
sudo gedit .bashrc
```

- Dodati naredbu *source /opt/ros/humble/setup.bash* na kraj datoteke

```
...
source /opt/ros/humble/setup.bash
```



Instalacija *VS Code*-a

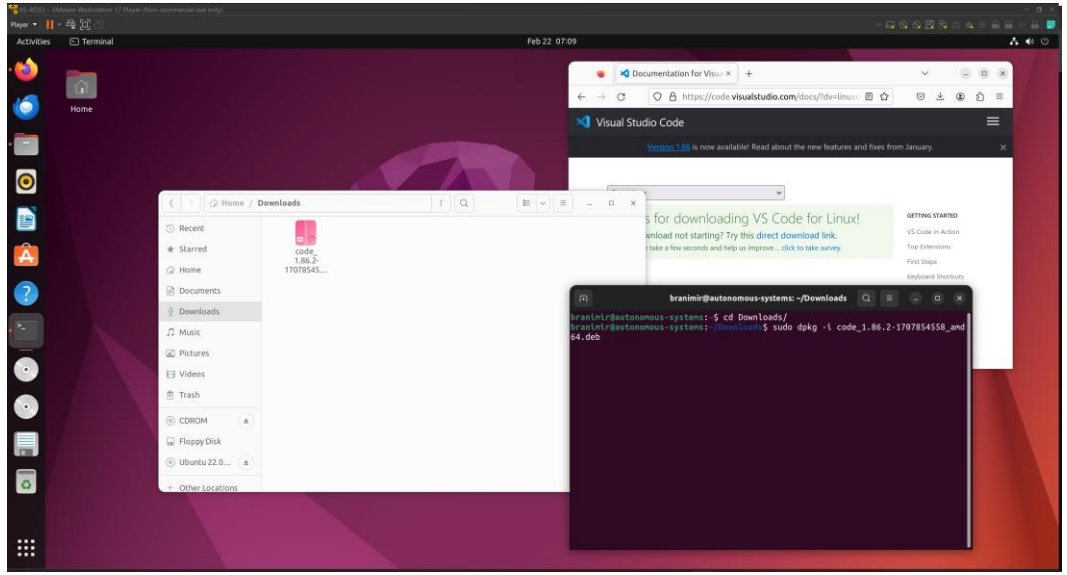
- *Visual Studio Code* potrebno je preuzeti s [linka]

- Potom je potrebno iz Terminala ući u folder *Downloads*

```
cd Downloads/
```

- I pokrenuti instalaciju

```
sudo dpkg -i code_1.86.2-1707854558_amd64.deb
```

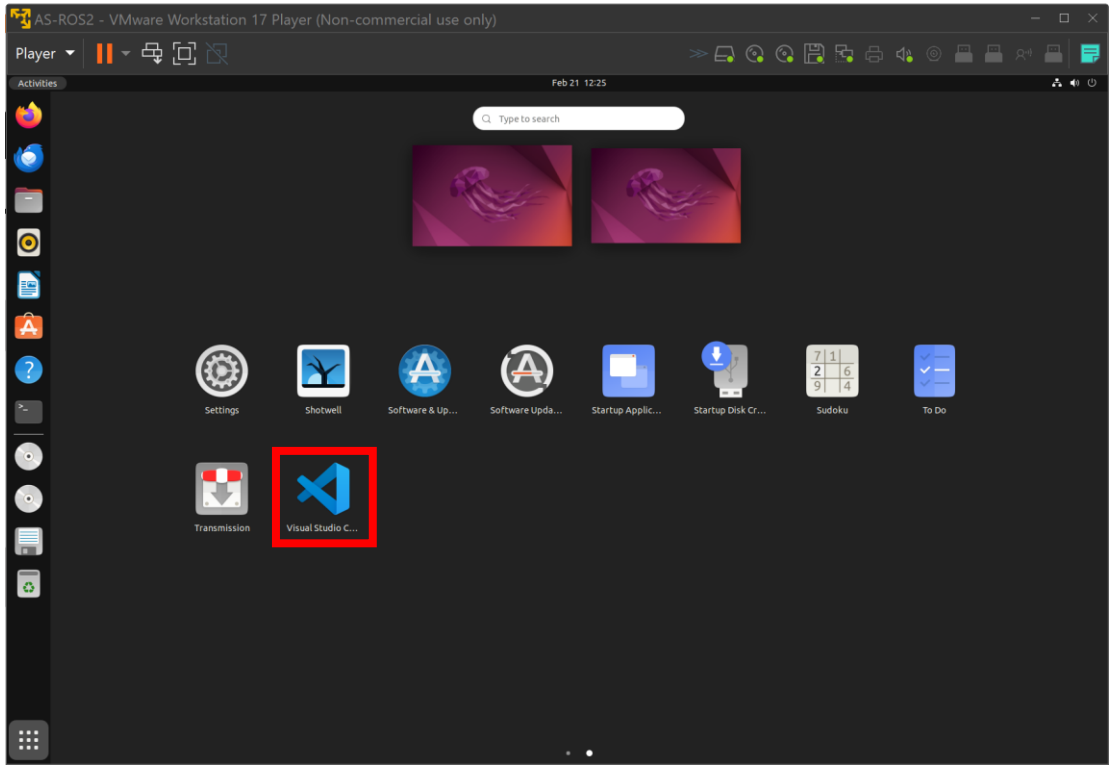


Pokretanje *VS Code*-a

- *Visual Studio Code* je moguće pokrenuti iz *Terminala* naredbom

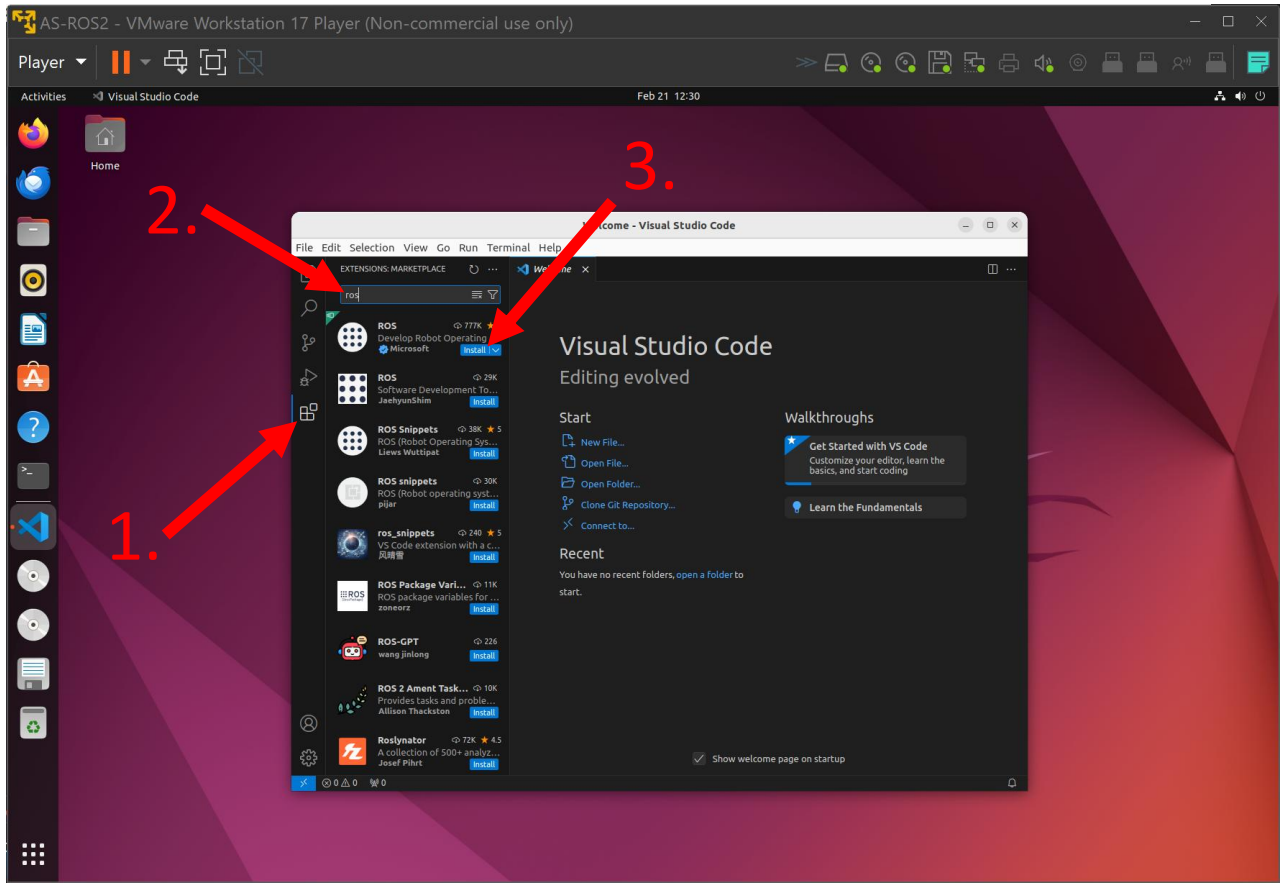
```
code
```

- Ili potražiti u instaliranim paketima



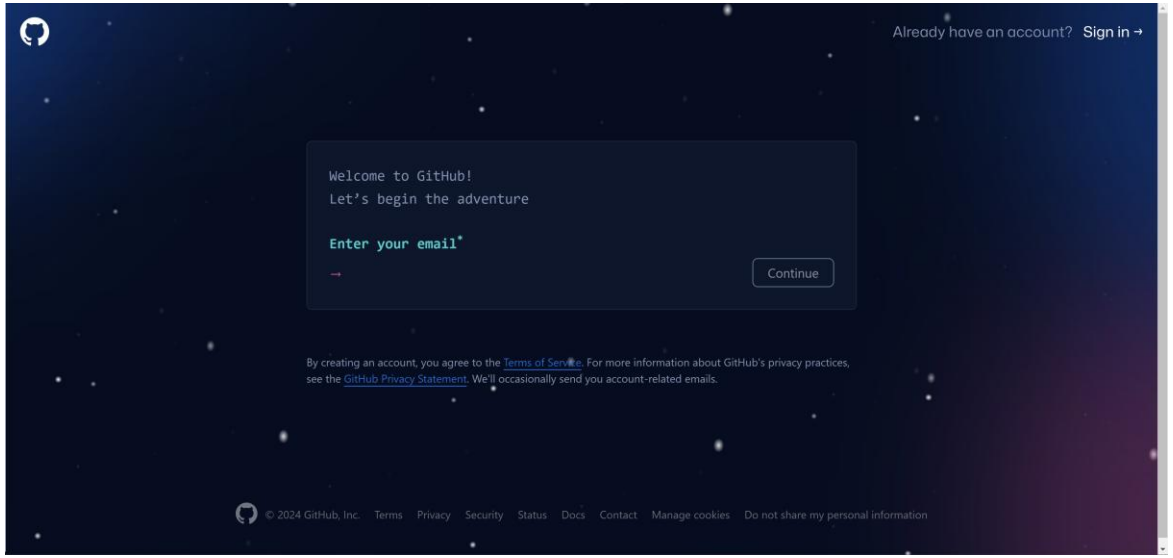
Instalacija proširenja

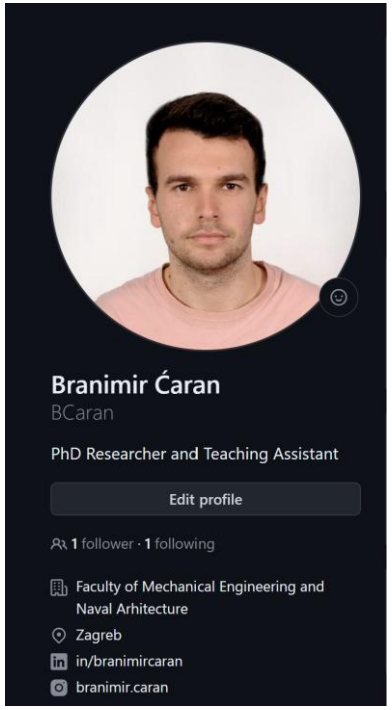
- Unutar *VSCode*-a je potrebno instalirati proširenje za *ROS*

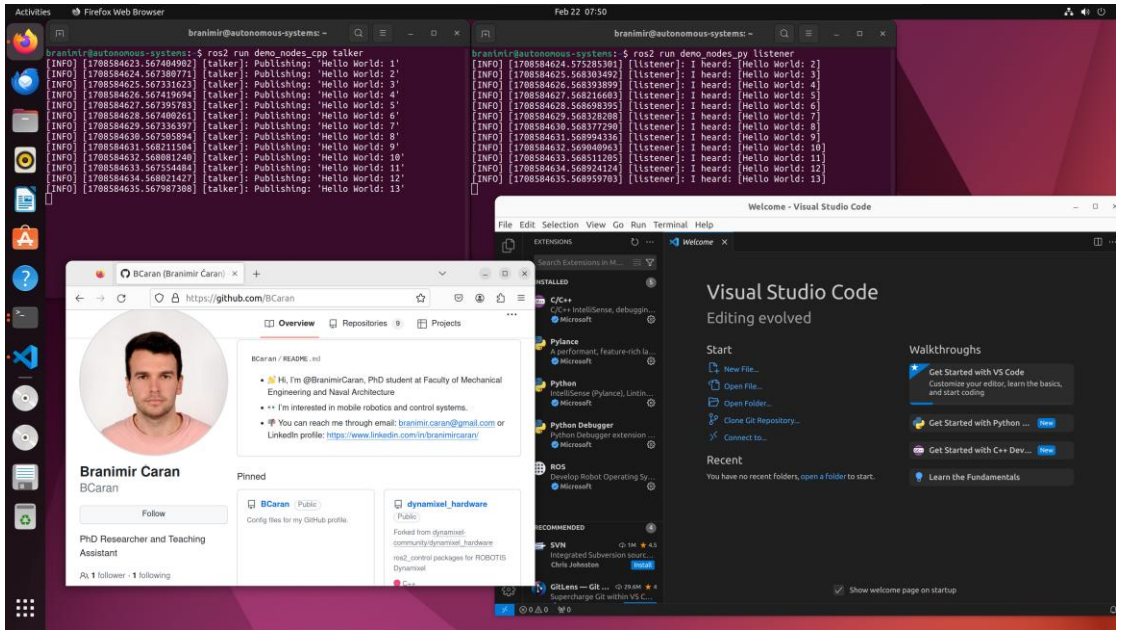


Registracija na *github.com*

- Zbog verzioniranja programa i praćenja napretka, za potrebe kolegija potrebno je napraviti račun na *github-u*
- Prilikom izrade *github* računa, potrebno je koristiti prezime i jedno ili više slova imena
- Preporuka je staviti sliku





Zadaća

- Instalirati virtualnu mašinu s *Linux Ubuntu 22.04* OS u *VMWare* programu
- Instalirati *ROS2 Humble* i pokrenuti primjere
- Instalirati *VSCode* i *ROS* proširenje
- Napraviti *github* račun
- Zadaća se predaje kao *screenshot* radne površine virtualne mašine (s pokrenutim primjerima, *VSCode*-om i profilom na *github.com*) preko e-učenja
- Rok je 31.03.2025. u 23:59